

Comprehensive Area Models Document for SKY130

1. MOSFET Devices

MOSFET area is modeled using a bounding-box approach rather than a flat linear sum. The bounding box Height depends linearly on Width (W), and the bounding box Width depends on Finger Length (L) and Number of Fingers (nf). Area is given by the cross-product:

$$\text{Area} = (\text{ah} * \text{W} + \text{bh}) * (\text{aw} * \text{L} * \text{nf} + \text{bw} * \text{nf} + \text{cw})$$

Device: nfet_01v8

Parameters: ah=1.0000, bh=2.1000, aw=0.9971, bw=0.3176, cw=1.6414

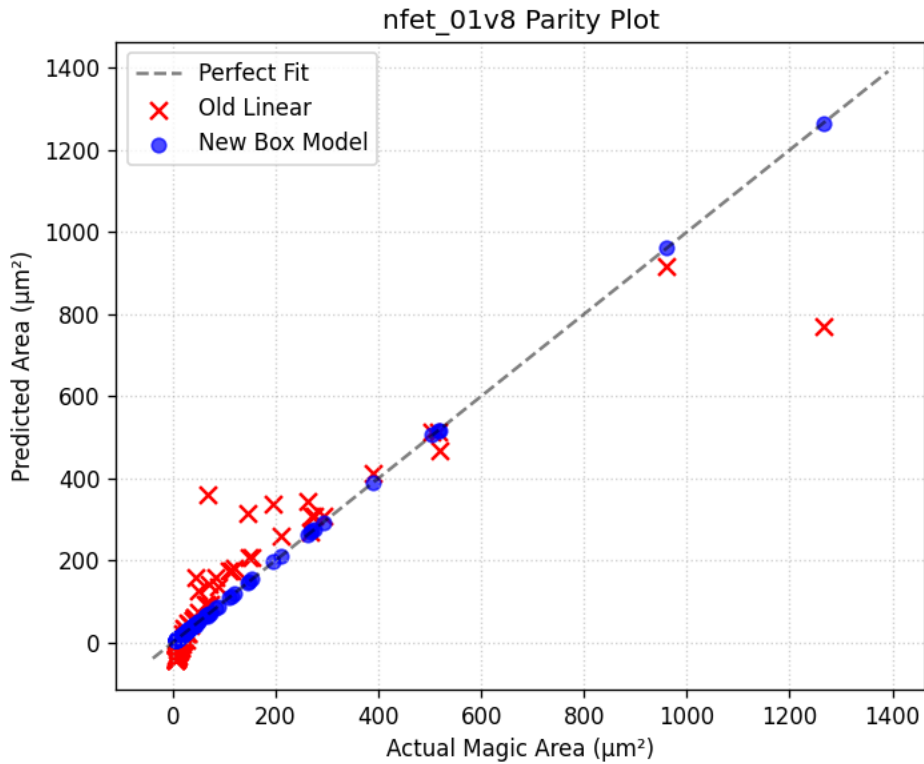
Height Model: $h_{\text{um}} = 1.0000 * \text{W} + 2.1000$

Width Model: $w_{\text{um}} = 0.9971 * (\text{L} * \text{nf}) + 0.3176 * \text{nf} + 1.6414$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	5.32	-43.38 (915.9%)	5.31 (0.1%)
W=0.42 L=0.15 nf=2	6.63	-41.26 (722.5%)	6.49 (2.1%)
W=0.42 L=0.15 nf=4	9.05	-37.00 (509.0%)	8.85 (2.2%)
W=0.42 L=0.5 nf=1	6.20	-32.40 (622.7%)	6.19 (0.1%)
W=0.42 L=1.0 nf=1	7.46	-16.72 (324.1%)	7.45 (0.1%)
W=0.45 L=0.15 nf=1	5.38	-43.23 (903.5%)	5.38 (0.1%)
W=0.45 L=0.15 nf=2	6.71	-40.95 (710.6%)	6.57 (2.1%)
W=0.45 L=0.15 nf=4	9.15	-36.39 (497.5%)	8.95 (2.2%)
W=0.6 L=0.15 nf=1	5.70	-42.47 (845.5%)	5.69 (0.1%)
W=0.6 L=0.15 nf=2	7.10	-39.43 (655.3%)	6.95 (2.1%)
W=0.6 L=0.15 nf=4	9.69	-33.35 (444.1%)	9.48 (2.2%)
W=1.0 L=0.15 nf=1	6.54	-40.45 (718.3%)	6.54 (0.1%)
W=1.0 L=0.15 nf=2	8.15	-35.38 (533.9%)	7.98 (2.1%)
W=1.0 L=0.5 nf=1	7.63	-29.46 (486.4%)	7.62 (0.1%)
W=1.0 L=0.5 nf=2	10.07	-24.40 (342.2%)	10.15 (0.7%)
W=1.0 L=0.5 nf=4	14.97	-14.26 (195.3%)	15.21 (1.6%)
W=1.0 L=2.0 nf=2	19.38	22.66 (17.0%)	19.42 (0.2%)
W=1.2 L=0.15 nf=1	6.96	-39.43 (666.3%)	6.96 (0.1%)
W=1.2 L=0.15 nf=2	8.68	-33.35 (484.3%)	8.50 (2.1%)
W=1.2 L=0.15 nf=4	11.85	-21.19 (278.9%)	11.58 (2.2%)
W=1.5 L=0.5 nf=1	8.86	-26.93 (404.1%)	8.85 (0.1%)
W=1.5 L=0.5 nf=2	11.70	-19.33 (265.2%)	11.79 (0.7%)
W=1.5 L=0.5 nf=4	17.39	-4.13 (123.8%)	17.66 (1.6%)
W=2.0 L=0.15 nf=4	14.72	-4.98 (133.8%)	14.39 (2.2%)
W=2.0 L=0.35 nf=2	12.10	-18.97 (256.8%)	12.20 (0.8%)
W=2.0 L=0.5 nf=1	10.09	-24.40 (341.9%)	10.08 (0.1%)
W=2.0 L=0.5 nf=2	13.32	-14.26 (207.0%)	13.42 (0.7%)

W=2.0 L=0.5 nf=4	19.80	6.00 (69.7%)	20.11 (1.6%)
W=2.0 L=1.0 nf=1	12.14	-8.71 (171.8%)	12.12 (0.1%)
W=2.0 L=1.0 nf=2	17.43	1.42 (91.8%)	17.51 (0.5%)
W=2.0 L=1.0 nf=4	28.00	21.69 (22.5%)	28.29 (1.0%)
W=2.0 L=5.0 nf=2	50.23	126.92 (152.7%)	50.22 (0.0%)
W=3.0 L=1.0 nf=1	15.10	-3.64 (124.1%)	15.08 (0.1%)
W=3.0 L=1.0 nf=2	21.68	11.56 (46.7%)	21.78 (0.5%)
W=3.0 L=1.0 nf=4	34.83	41.96 (20.5%)	35.19 (1.0%)
W=4.0 L=0.15 nf=4	21.90	35.56 (62.4%)	21.41 (2.2%)
W=4.0 L=0.5 nf=1	15.01	-14.26 (195.1%)	14.99 (0.1%)
W=4.0 L=0.5 nf=2	19.82	6.00 (69.7%)	19.97 (0.7%)
W=4.0 L=0.5 nf=4	29.46	46.54 (58.0%)	29.93 (1.6%)
W=4.0 L=1.0 nf=1	18.06	1.42 (92.1%)	18.03 (0.1%)
W=4.0 L=1.0 nf=2	25.93	21.69 (16.3%)	26.05 (0.5%)
W=4.0 L=1.0 nf=4	41.66	62.23 (49.4%)	42.09 (1.0%)
W=4.0 L=2.0 nf=1	24.16	32.80 (35.8%)	24.11 (0.2%)
W=4.0 L=2.0 nf=2	38.12	53.06 (39.2%)	38.22 (0.2%)
W=4.0 L=2.0 nf=4	66.06	93.60 (41.7%)	66.42 (0.5%)
W=4.0 L=5.0 nf=8	268.34	268.79 (0.2%)	268.80 (0.2%)
W=4.0 L=10.0 nf=4	261.26	344.59 (31.9%)	261.05 (0.1%)
W=6.0 L=2.0 nf=1	32.08	42.93 (33.8%)	32.02 (0.2%)
W=6.0 L=2.0 nf=2	50.62	73.33 (44.9%)	50.75 (0.2%)
W=6.0 L=2.0 nf=4	87.72	134.14 (52.9%)	88.20 (0.5%)
W=8.0 L=1.0 nf=1	29.90	21.69 (27.4%)	29.86 (0.1%)
W=8.0 L=1.0 nf=2	42.92	62.23 (45.0%)	43.14 (0.5%)
W=8.0 L=1.0 nf=4	68.98	143.30 (107.7%)	69.69 (1.0%)
W=8.0 L=2.0 nf=1	40.00	53.06 (32.7%)	39.93 (0.2%)
W=8.0 L=2.0 nf=2	63.12	93.60 (48.3%)	63.28 (0.2%)
W=8.0 L=2.0 nf=4	109.38	174.67 (59.7%)	109.97 (0.5%)
W=10.0 L=0.15 nf=1	25.53	5.16 (79.8%)	25.51 (0.1%)
W=10.0 L=0.15 nf=4	43.44	157.16 (261.8%)	42.47 (2.2%)
W=10.0 L=0.15 nf=8	66.67	359.84 (439.7%)	65.08 (2.4%)
W=10.0 L=5.0 nf=1	84.22	157.32 (86.8%)	84.03 (0.2%)
W=10.0 L=5.0 nf=2	148.22	207.99 (40.3%)	148.20 (0.0%)
W=10.0 L=5.0 nf=4	276.24	309.33 (12.0%)	276.53 (0.1%)
W=10.0 L=10.0 nf=1	144.72	314.19 (117.1%)	144.35 (0.3%)
W=10.0 L=10.0 nf=4	518.24	466.20 (10.0%)	517.83 (0.1%)
W=10.0 L=10.0 nf=10	1265.30	770.21 (39.1%)	1264.78 (0.0%)
W=15.0 L=5.0 nf=1	119.02	182.65 (53.5%)	118.75 (0.2%)
W=15.0 L=5.0 nf=2	209.47	258.66 (23.5%)	209.43 (0.0%)
W=15.0 L=5.0 nf=4	390.39	410.67 (5.2%)	390.80 (0.1%)
W=16.0 L=2.0 nf=1	71.68	93.60 (30.6%)	71.55 (0.2%)
W=16.0 L=2.0 nf=2	113.12	174.67 (54.4%)	113.40 (0.2%)
W=16.0 L=2.0 nf=4	196.02	336.81 (71.8%)	197.08 (0.5%)
W=20.0 L=5.0 nf=1	153.82	207.99 (35.2%)	153.47 (0.2%)
W=20.0 L=5.0 nf=2	270.73	309.33 (14.3%)	270.67 (0.0%)
W=20.0 L=5.0 nf=4	504.54	512.00 (1.5%)	505.07 (0.1%)

W=40.0 L=5.0 nf=1	293.02	309.33 (5.6%)	292.36 (0.2%)
W=40.0 L=5.0 nf=2	515.73	512.00 (0.7%)	515.62 (0.0%)
W=40.0 L=5.0 nf=4	961.14	917.36 (4.6%)	962.14 (0.1%)



Device: pfet_01v8

Parameters: ah=1.0000, bh=2.1900, aw=0.9971, bw=0.3176, cw=1.6414

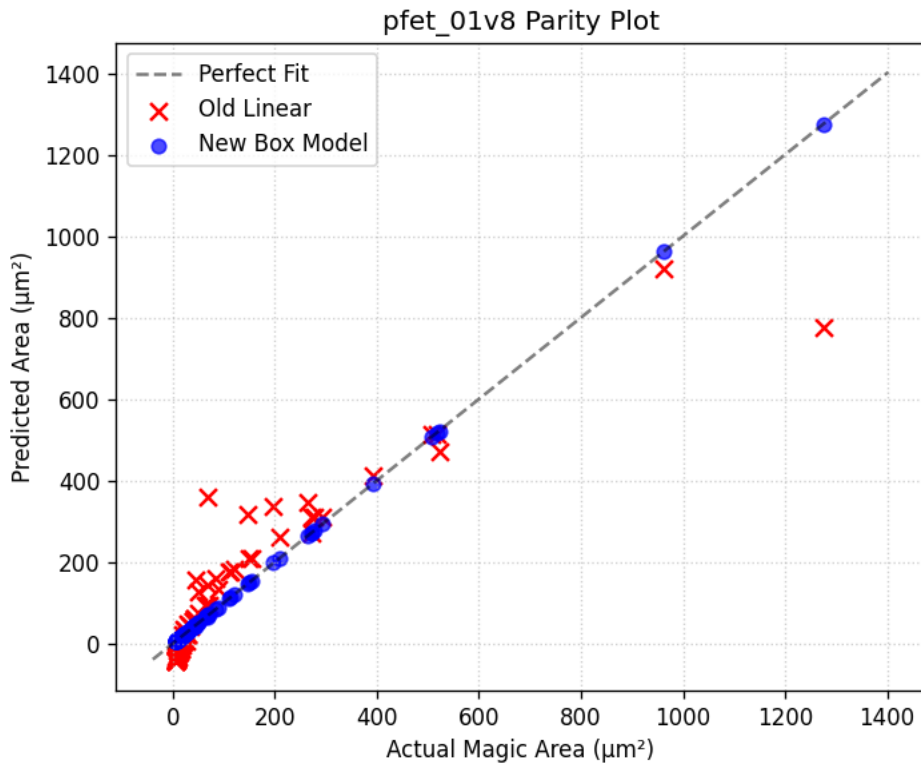
Height Model: $h_{um} = 1.0000 \cdot W + 2.1900$

Width Model: $w_{um} = 0.9971 \cdot (L \cdot nf) + 0.3176 \cdot nf + 1.6414$

Dimensions	Actual (µm²)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	5.51	-43.39 (887.8%)	5.50 (0.1%)
W=0.42 L=0.15 nf=2	6.86	-41.25 (701.0%)	6.72 (2.1%)
W=0.42 L=0.15 nf=4	9.37	-36.99 (494.7%)	9.16 (2.2%)
W=0.42 L=0.5 nf=1	6.42	-32.30 (603.1%)	6.41 (0.1%)
W=0.42 L=1.0 nf=1	7.73	-16.46 (313.1%)	7.72 (0.1%)
W=0.45 L=0.15 nf=1	5.57	-43.23 (876.1%)	5.57 (0.1%)
W=0.45 L=0.15 nf=2	6.94	-40.95 (689.8%)	6.80 (2.1%)
W=0.45 L=0.15 nf=4	9.48	-36.38 (483.8%)	9.27 (2.2%)
W=0.6 L=0.15 nf=1	5.89	-42.47 (821.5%)	5.88 (0.1%)
W=0.6 L=0.15 nf=2	7.34	-39.43 (637.3%)	7.19 (2.1%)
W=0.6 L=0.15 nf=4	10.02	-33.33 (432.8%)	9.79 (2.2%)
W=1.0 L=0.15 nf=1	6.73	-40.44 (700.8%)	6.73 (0.1%)
W=1.0 L=0.15 nf=2	8.39	-35.36 (521.5%)	8.22 (2.1%)

W=1.0 L=0.5 nf=1	7.85	-29.35 (474.1%)	7.84 (0.1%)
W=1.0 L=0.5 nf=2	10.37	-24.27 (334.1%)	10.44 (0.7%)
W=1.0 L=0.5 nf=4	15.41	-14.12 (191.6%)	15.65 (1.6%)
W=1.0 L=2.0 nf=2	19.94	23.24 (16.6%)	19.99 (0.2%)
W=1.2 L=0.15 nf=1	7.15	-39.43 (651.2%)	7.15 (0.1%)
W=1.2 L=0.15 nf=2	8.92	-33.33 (473.8%)	8.73 (2.1%)
W=1.2 L=0.15 nf=4	12.17	-21.14 (273.7%)	11.90 (2.2%)
W=1.5 L=0.5 nf=1	9.08	-26.81 (395.4%)	9.07 (0.1%)
W=1.5 L=0.5 nf=2	11.99	-19.20 (260.1%)	12.08 (0.7%)
W=1.5 L=0.5 nf=4	17.82	-3.96 (122.2%)	18.10 (1.6%)
W=2.0 L=0.15 nf=4	15.04	-4.89 (132.5%)	14.71 (2.2%)
W=2.0 L=0.35 nf=2	12.36	-18.87 (252.6%)	12.46 (0.8%)
W=2.0 L=0.5 nf=1	10.31	-24.27 (335.5%)	10.30 (0.1%)
W=2.0 L=0.5 nf=2	13.62	-14.12 (203.7%)	13.72 (0.7%)
W=2.0 L=0.5 nf=4	20.24	6.20 (69.4%)	20.56 (1.6%)
W=2.0 L=1.0 nf=1	12.40	-8.44 (168.0%)	12.39 (0.1%)
W=2.0 L=1.0 nf=2	17.81	1.72 (90.3%)	17.89 (0.5%)
W=2.0 L=1.0 nf=4	28.62	22.04 (23.0%)	28.91 (1.0%)
W=2.0 L=5.0 nf=2	51.33	128.43 (150.2%)	51.32 (0.0%)
W=3.0 L=1.0 nf=1	15.36	-3.36 (121.8%)	15.34 (0.1%)
W=3.0 L=1.0 nf=2	22.06	11.88 (46.1%)	22.17 (0.5%)
W=3.0 L=1.0 nf=4	35.45	42.36 (19.5%)	35.81 (1.0%)
W=4.0 L=0.15 nf=4	22.22	35.75 (60.9%)	21.73 (2.2%)
W=4.0 L=0.5 nf=1	15.23	-14.12 (192.7%)	15.21 (0.1%)
W=4.0 L=0.5 nf=2	20.12	6.20 (69.2%)	20.26 (0.7%)
W=4.0 L=0.5 nf=4	29.90	46.83 (56.6%)	30.37 (1.6%)
W=4.0 L=1.0 nf=1	18.32	1.72 (90.6%)	18.30 (0.1%)
W=4.0 L=1.0 nf=2	26.31	22.04 (16.2%)	26.44 (0.5%)
W=4.0 L=1.0 nf=4	42.28	62.67 (48.2%)	42.71 (1.0%)
W=4.0 L=2.0 nf=1	24.51	33.40 (36.3%)	24.47 (0.2%)
W=4.0 L=2.0 nf=2	38.69	53.72 (38.8%)	38.78 (0.2%)
W=4.0 L=2.0 nf=4	67.04	94.35 (40.7%)	67.40 (0.5%)
W=4.0 L=5.0 nf=8	272.30	270.65 (0.6%)	272.77 (0.2%)
W=4.0 L=10.0 nf=4	265.12	347.77 (31.2%)	264.91 (0.1%)
W=6.0 L=2.0 nf=1	32.43	43.56 (34.3%)	32.38 (0.2%)
W=6.0 L=2.0 nf=2	51.19	74.03 (44.6%)	51.31 (0.2%)
W=6.0 L=2.0 nf=4	88.70	134.98 (52.2%)	89.18 (0.5%)
W=8.0 L=1.0 nf=1	30.16	22.04 (26.9%)	30.12 (0.1%)
W=8.0 L=1.0 nf=2	43.31	62.67 (44.7%)	43.52 (0.5%)
W=8.0 L=1.0 nf=4	69.60	143.94 (106.8%)	70.31 (1.0%)
W=8.0 L=2.0 nf=1	40.35	53.72 (33.1%)	40.28 (0.2%)
W=8.0 L=2.0 nf=2	63.69	94.35 (48.1%)	63.84 (0.2%)
W=8.0 L=2.0 nf=4	110.36	175.62 (59.1%)	110.95 (0.5%)
W=10.0 L=0.15 nf=1	25.72	5.27 (79.5%)	25.70 (0.1%)
W=10.0 L=0.15 nf=4	43.76	157.64 (260.2%)	42.79 (2.2%)
W=10.0 L=0.15 nf=8	67.17	360.81 (437.2%)	65.57 (2.4%)
W=10.0 L=5.0 nf=1	84.84	158.91 (87.3%)	84.65 (0.2%)
W=10.0 L=5.0 nf=2	149.33	209.70 (40.4%)	149.30 (0.0%)

W=10.0 L=5.0 nf=4	278.30	311.28 (11.9%)	278.59 (0.1%)
W=10.0 L=10.0 nf=1	145.79	317.30 (117.6%)	145.43 (0.3%)
W=10.0 L=10.0 nf=4	522.10	469.67 (10.0%)	521.68 (0.1%)
W=10.0 L=10.0 nf=10	1274.71	774.42 (39.2%)	1274.19 (0.0%)
W=15.0 L=5.0 nf=1	119.64	184.30 (54.0%)	119.38 (0.2%)
W=15.0 L=5.0 nf=2	210.58	260.49 (23.7%)	210.54 (0.0%)
W=15.0 L=5.0 nf=4	392.45	412.87 (5.2%)	392.86 (0.1%)
W=16.0 L=2.0 nf=1	72.03	94.35 (31.0%)	71.91 (0.2%)
W=16.0 L=2.0 nf=2	113.69	175.62 (54.5%)	113.96 (0.2%)
W=16.0 L=2.0 nf=4	197.00	338.15 (71.7%)	198.06 (0.5%)
W=20.0 L=5.0 nf=1	154.44	209.70 (35.8%)	154.10 (0.2%)
W=20.0 L=5.0 nf=2	271.83	311.28 (14.5%)	271.77 (0.0%)
W=20.0 L=5.0 nf=4	506.60	514.45 (1.5%)	507.13 (0.1%)
W=40.0 L=5.0 nf=1	293.64	311.28 (6.0%)	292.99 (0.2%)
W=40.0 L=5.0 nf=2	516.83	514.45 (0.5%)	516.73 (0.0%)
W=40.0 L=5.0 nf=4	963.20	920.78 (4.4%)	964.20 (0.1%)



Device: nfet_01v8_lvt

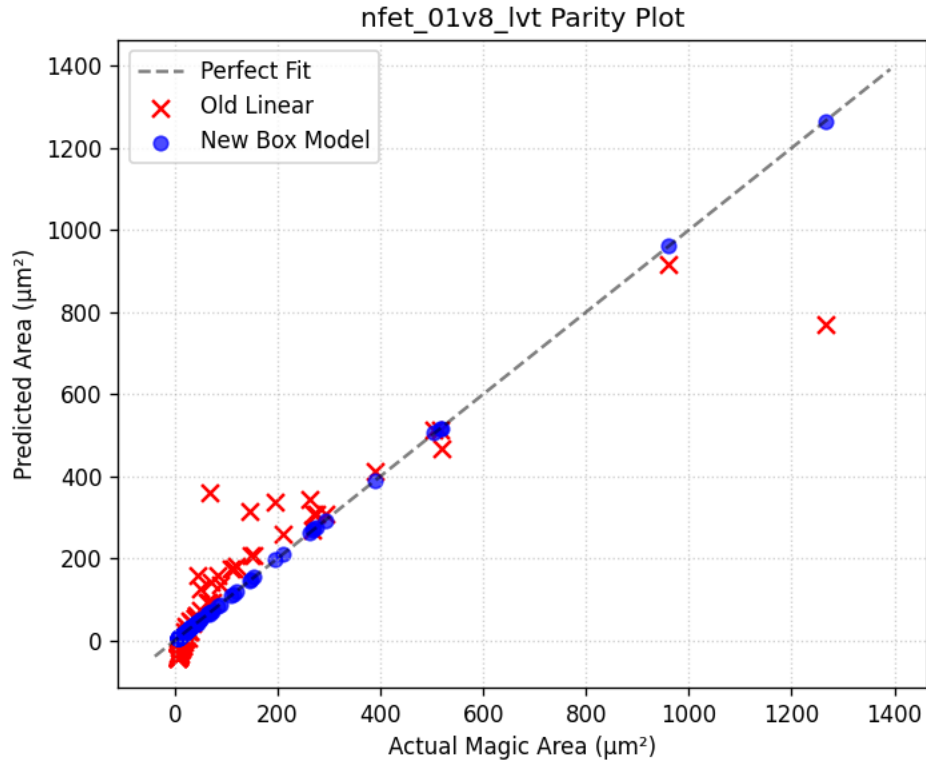
Parameters: ah=1.0000, bh=2.1000, aw=0.9971, bw=0.3176, cw=1.6414

Height Model: $h_{\text{um}} = 1.0000 \cdot W + 2.1000$

Width Model: $w_{\text{um}} = 0.9971 \cdot (L \cdot \text{nf}) + 0.3176 \cdot \text{nf} + 1.6414$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	5.32	-43.38 (915.9%)	5.31 (0.1%)
W=0.42 L=0.15 nf=2	6.63	-41.26 (722.5%)	6.49 (2.1%)
W=0.42 L=0.15 nf=4	9.05	-37.00 (509.0%)	8.85 (2.2%)
W=0.42 L=0.5 nf=1	6.20	-32.40 (622.7%)	6.19 (0.1%)
W=0.42 L=1.0 nf=1	7.46	-16.72 (324.1%)	7.45 (0.1%)
W=0.45 L=0.15 nf=1	5.38	-43.23 (903.5%)	5.38 (0.1%)
W=0.45 L=0.15 nf=2	6.71	-40.95 (710.6%)	6.57 (2.1%)
W=0.45 L=0.15 nf=4	9.15	-36.39 (497.5%)	8.95 (2.2%)
W=0.6 L=0.15 nf=1	5.70	-42.47 (845.5%)	5.69 (0.1%)
W=0.6 L=0.15 nf=2	7.10	-39.43 (655.3%)	6.95 (2.1%)
W=0.6 L=0.15 nf=4	9.69	-33.35 (444.1%)	9.48 (2.2%)
W=1.0 L=0.15 nf=1	6.54	-40.45 (718.3%)	6.54 (0.1%)
W=1.0 L=0.15 nf=2	8.15	-35.38 (533.9%)	7.98 (2.1%)
W=1.0 L=0.5 nf=1	7.63	-29.46 (486.4%)	7.62 (0.1%)
W=1.0 L=0.5 nf=2	10.07	-24.40 (342.2%)	10.15 (0.7%)
W=1.0 L=0.5 nf=4	14.97	-14.26 (195.3%)	15.21 (1.6%)
W=1.0 L=2.0 nf=2	19.38	22.66 (17.0%)	19.42 (0.2%)
W=1.2 L=0.15 nf=1	6.96	-39.43 (666.3%)	6.96 (0.1%)
W=1.2 L=0.15 nf=2	8.68	-33.35 (484.3%)	8.50 (2.1%)
W=1.2 L=0.15 nf=4	11.85	-21.19 (278.9%)	11.58 (2.2%)
W=1.5 L=0.5 nf=1	8.86	-26.93 (404.1%)	8.85 (0.1%)
W=1.5 L=0.5 nf=2	11.70	-19.33 (265.2%)	11.79 (0.7%)
W=1.5 L=0.5 nf=4	17.39	-4.13 (123.8%)	17.66 (1.6%)
W=2.0 L=0.15 nf=4	14.72	-4.98 (133.8%)	14.39 (2.2%)
W=2.0 L=0.35 nf=2	12.10	-18.97 (256.8%)	12.20 (0.8%)
W=2.0 L=0.5 nf=1	10.09	-24.40 (341.9%)	10.08 (0.1%)
W=2.0 L=0.5 nf=2	13.32	-14.26 (207.0%)	13.42 (0.7%)
W=2.0 L=0.5 nf=4	19.80	6.00 (69.7%)	20.11 (1.6%)
W=2.0 L=1.0 nf=1	12.14	-8.71 (171.8%)	12.12 (0.1%)
W=2.0 L=1.0 nf=2	17.43	1.42 (91.8%)	17.51 (0.5%)
W=2.0 L=1.0 nf=4	28.00	21.69 (22.5%)	28.29 (1.0%)
W=2.0 L=5.0 nf=2	50.23	126.92 (152.7%)	50.22 (0.0%)
W=3.0 L=1.0 nf=1	15.10	-3.64 (124.1%)	15.08 (0.1%)
W=3.0 L=1.0 nf=2	21.68	11.56 (46.7%)	21.78 (0.5%)
W=3.0 L=1.0 nf=4	34.83	41.96 (20.5%)	35.19 (1.0%)
W=4.0 L=0.15 nf=4	21.90	35.56 (62.4%)	21.41 (2.2%)
W=4.0 L=0.5 nf=1	15.01	-14.26 (195.1%)	14.99 (0.1%)
W=4.0 L=0.5 nf=2	19.82	6.00 (69.7%)	19.97 (0.7%)
W=4.0 L=0.5 nf=4	29.46	46.54 (58.0%)	29.93 (1.6%)
W=4.0 L=1.0 nf=1	18.06	1.42 (92.1%)	18.03 (0.1%)
W=4.0 L=1.0 nf=2	25.93	21.69 (16.3%)	26.05 (0.5%)
W=4.0 L=1.0 nf=4	41.66	62.23 (49.4%)	42.09 (1.0%)
W=4.0 L=2.0 nf=1	24.16	32.80 (35.8%)	24.11 (0.2%)
W=4.0 L=2.0 nf=2	38.12	53.06 (39.2%)	38.22 (0.2%)
W=4.0 L=2.0 nf=4	66.06	93.60 (41.7%)	66.42 (0.5%)
W=4.0 L=5.0 nf=8	268.34	268.79 (0.2%)	268.80 (0.2%)
W=4.0 L=10.0 nf=4	261.26	344.59 (31.9%)	261.05 (0.1%)

W=6.0 L=2.0 nf=1	32.08	42.93 (33.8%)	32.02 (0.2%)
W=6.0 L=2.0 nf=2	50.62	73.33 (44.9%)	50.75 (0.2%)
W=6.0 L=2.0 nf=4	87.72	134.14 (52.9%)	88.20 (0.5%)
W=8.0 L=1.0 nf=1	29.90	21.69 (27.4%)	29.86 (0.1%)
W=8.0 L=1.0 nf=2	42.92	62.23 (45.0%)	43.14 (0.5%)
W=8.0 L=1.0 nf=4	68.98	143.30 (107.7%)	69.69 (1.0%)
W=8.0 L=2.0 nf=1	40.00	53.06 (32.7%)	39.93 (0.2%)
W=8.0 L=2.0 nf=2	63.12	93.60 (48.3%)	63.28 (0.2%)
W=8.0 L=2.0 nf=4	109.38	174.67 (59.7%)	109.97 (0.5%)
W=10.0 L=0.15 nf=1	25.53	5.16 (79.8%)	25.51 (0.1%)
W=10.0 L=0.15 nf=4	43.44	157.16 (261.8%)	42.47 (2.2%)
W=10.0 L=0.15 nf=8	66.67	359.84 (439.7%)	65.08 (2.4%)
W=10.0 L=5.0 nf=1	84.22	157.32 (86.8%)	84.03 (0.2%)
W=10.0 L=5.0 nf=2	148.22	207.99 (40.3%)	148.20 (0.0%)
W=10.0 L=5.0 nf=4	276.24	309.33 (12.0%)	276.53 (0.1%)
W=10.0 L=10.0 nf=1	144.72	314.19 (117.1%)	144.35 (0.3%)
W=10.0 L=10.0 nf=4	518.24	466.20 (10.0%)	517.83 (0.1%)
W=10.0 L=10.0 nf=10	1265.30	770.21 (39.1%)	1264.78 (0.0%)
W=15.0 L=5.0 nf=1	119.02	182.65 (53.5%)	118.75 (0.2%)
W=15.0 L=5.0 nf=2	209.47	258.66 (23.5%)	209.43 (0.0%)
W=15.0 L=5.0 nf=4	390.39	410.67 (5.2%)	390.80 (0.1%)
W=16.0 L=2.0 nf=1	71.68	93.60 (30.6%)	71.55 (0.2%)
W=16.0 L=2.0 nf=2	113.12	174.67 (54.4%)	113.40 (0.2%)
W=16.0 L=2.0 nf=4	196.02	336.81 (71.8%)	197.08 (0.5%)
W=20.0 L=5.0 nf=1	153.82	207.99 (35.2%)	153.47 (0.2%)
W=20.0 L=5.0 nf=2	270.73	309.33 (14.3%)	270.67 (0.0%)
W=20.0 L=5.0 nf=4	504.54	512.00 (1.5%)	505.07 (0.1%)
W=40.0 L=5.0 nf=1	293.02	309.33 (5.6%)	292.36 (0.2%)
W=40.0 L=5.0 nf=2	515.73	512.00 (0.7%)	515.62 (0.0%)
W=40.0 L=5.0 nf=4	961.14	917.36 (4.6%)	962.14 (0.1%)



Device: pfet_01v8_hvt

Parameters: ah=1.0000, bh=2.1900, aw=0.9971, bw=0.3176, cw=1.6414

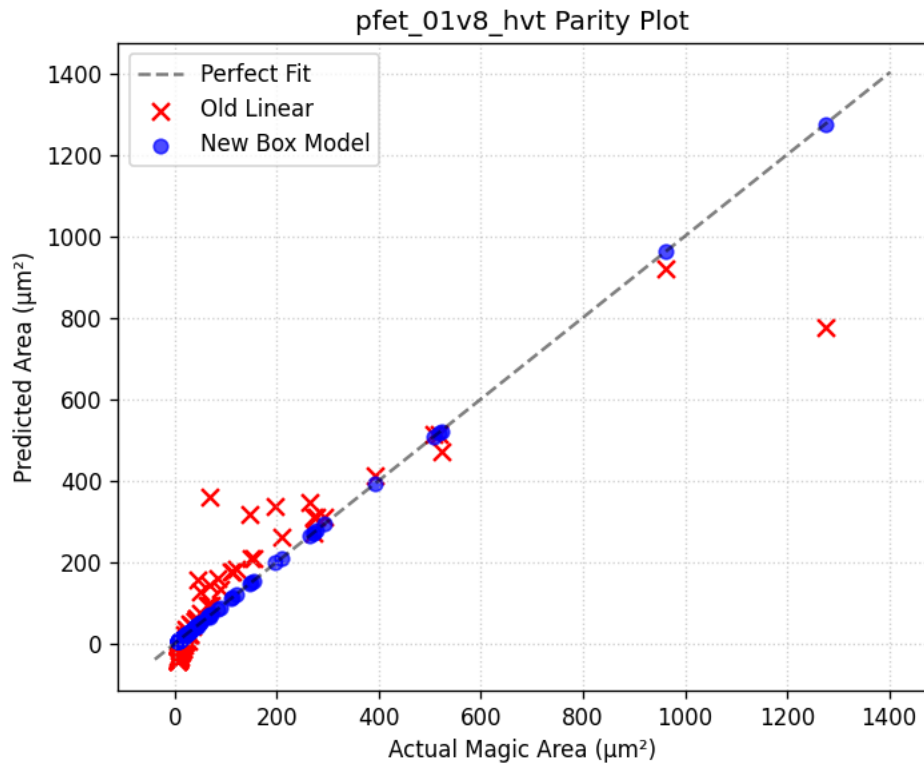
Height Model: $h_{um} = 1.0000 \cdot W + 2.1900$

Width Model: $w_{um} = 0.9971 \cdot (L \cdot nf) + 0.3176 \cdot nf + 1.6414$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	5.51	-43.39 (887.8%)	5.50 (0.1%)
W=0.42 L=0.15 nf=2	6.86	-41.25 (701.0%)	6.72 (2.1%)
W=0.42 L=0.15 nf=4	9.37	-36.99 (494.7%)	9.16 (2.2%)
W=0.42 L=0.5 nf=1	6.42	-32.30 (603.1%)	6.41 (0.1%)
W=0.42 L=1.0 nf=1	7.73	-16.46 (313.1%)	7.72 (0.1%)
W=0.45 L=0.15 nf=1	5.57	-43.23 (876.1%)	5.57 (0.1%)
W=0.45 L=0.15 nf=2	6.94	-40.95 (689.8%)	6.80 (2.1%)
W=0.45 L=0.15 nf=4	9.48	-36.38 (483.8%)	9.27 (2.2%)
W=0.6 L=0.15 nf=1	5.89	-42.47 (821.5%)	5.88 (0.1%)
W=0.6 L=0.15 nf=2	7.34	-39.43 (637.3%)	7.19 (2.1%)
W=0.6 L=0.15 nf=4	10.02	-33.33 (432.8%)	9.79 (2.2%)
W=1.0 L=0.15 nf=1	6.73	-40.44 (700.8%)	6.73 (0.1%)
W=1.0 L=0.15 nf=2	8.39	-35.36 (521.5%)	8.22 (2.1%)
W=1.0 L=0.5 nf=1	7.85	-29.35 (474.1%)	7.84 (0.1%)
W=1.0 L=0.5 nf=2	10.37	-24.27 (334.1%)	10.44 (0.7%)
W=1.0 L=0.5 nf=4	15.41	-14.12 (191.6%)	15.65 (1.6%)

W=1.0 L=2.0 nf=2	19.94	23.24 (16.6%)	19.99 (0.2%)
W=1.2 L=0.15 nf=1	7.15	-39.43 (651.2%)	7.15 (0.1%)
W=1.2 L=0.15 nf=2	8.92	-33.33 (473.8%)	8.73 (2.1%)
W=1.2 L=0.15 nf=4	12.17	-21.14 (273.7%)	11.90 (2.2%)
W=1.5 L=0.5 nf=1	9.08	-26.81 (395.4%)	9.07 (0.1%)
W=1.5 L=0.5 nf=2	11.99	-19.20 (260.1%)	12.08 (0.7%)
W=1.5 L=0.5 nf=4	17.82	-3.96 (122.2%)	18.10 (1.6%)
W=2.0 L=0.15 nf=4	15.04	-4.89 (132.5%)	14.71 (2.2%)
W=2.0 L=0.35 nf=2	12.36	-18.87 (252.6%)	12.46 (0.8%)
W=2.0 L=0.5 nf=1	10.31	-24.27 (335.5%)	10.30 (0.1%)
W=2.0 L=0.5 nf=2	13.62	-14.12 (203.7%)	13.72 (0.7%)
W=2.0 L=0.5 nf=4	20.24	6.20 (69.4%)	20.56 (1.6%)
W=2.0 L=1.0 nf=1	12.40	-8.44 (168.0%)	12.39 (0.1%)
W=2.0 L=1.0 nf=2	17.81	1.72 (90.3%)	17.89 (0.5%)
W=2.0 L=1.0 nf=4	28.62	22.04 (23.0%)	28.91 (1.0%)
W=2.0 L=5.0 nf=2	51.33	128.43 (150.2%)	51.32 (0.0%)
W=3.0 L=1.0 nf=1	15.36	-3.36 (121.8%)	15.34 (0.1%)
W=3.0 L=1.0 nf=2	22.06	11.88 (46.1%)	22.17 (0.5%)
W=3.0 L=1.0 nf=4	35.45	42.36 (19.5%)	35.81 (1.0%)
W=4.0 L=0.15 nf=4	22.22	35.75 (60.9%)	21.73 (2.2%)
W=4.0 L=0.5 nf=1	15.23	-14.12 (192.7%)	15.21 (0.1%)
W=4.0 L=0.5 nf=2	20.12	6.20 (69.2%)	20.26 (0.7%)
W=4.0 L=0.5 nf=4	29.90	46.83 (56.6%)	30.37 (1.6%)
W=4.0 L=1.0 nf=1	18.32	1.72 (90.6%)	18.30 (0.1%)
W=4.0 L=1.0 nf=2	26.31	22.04 (16.2%)	26.44 (0.5%)
W=4.0 L=1.0 nf=4	42.28	62.67 (48.2%)	42.71 (1.0%)
W=4.0 L=2.0 nf=1	24.51	33.40 (36.3%)	24.47 (0.2%)
W=4.0 L=2.0 nf=2	38.69	53.72 (38.8%)	38.78 (0.2%)
W=4.0 L=2.0 nf=4	67.04	94.35 (40.7%)	67.40 (0.5%)
W=4.0 L=5.0 nf=8	272.30	270.65 (0.6%)	272.77 (0.2%)
W=4.0 L=10.0 nf=4	265.12	347.77 (31.2%)	264.91 (0.1%)
W=6.0 L=2.0 nf=1	32.43	43.56 (34.3%)	32.38 (0.2%)
W=6.0 L=2.0 nf=2	51.19	74.03 (44.6%)	51.31 (0.2%)
W=6.0 L=2.0 nf=4	88.70	134.98 (52.2%)	89.18 (0.5%)
W=8.0 L=1.0 nf=1	30.16	22.04 (26.9%)	30.12 (0.1%)
W=8.0 L=1.0 nf=2	43.31	62.67 (44.7%)	43.52 (0.5%)
W=8.0 L=1.0 nf=4	69.60	143.94 (106.8%)	70.31 (1.0%)
W=8.0 L=2.0 nf=1	40.35	53.72 (33.1%)	40.28 (0.2%)
W=8.0 L=2.0 nf=2	63.69	94.35 (48.1%)	63.84 (0.2%)
W=8.0 L=2.0 nf=4	110.36	175.62 (59.1%)	110.95 (0.5%)
W=10.0 L=0.15 nf=1	25.72	5.27 (79.5%)	25.70 (0.1%)
W=10.0 L=0.15 nf=4	43.76	157.64 (260.2%)	42.79 (2.2%)
W=10.0 L=0.15 nf=8	67.17	360.81 (437.2%)	65.57 (2.4%)
W=10.0 L=5.0 nf=1	84.84	158.91 (87.3%)	84.65 (0.2%)
W=10.0 L=5.0 nf=2	149.33	209.70 (40.4%)	149.30 (0.0%)
W=10.0 L=5.0 nf=4	278.30	311.28 (11.9%)	278.59 (0.1%)
W=10.0 L=10.0 nf=1	145.79	317.30 (117.6%)	145.43 (0.3%)
W=10.0 L=10.0 nf=4	522.10	469.67 (10.0%)	521.68 (0.1%)

W=10.0 L=10.0 nf=10	1274.71	774.42 (39.2%)	1274.19 (0.0%)
W=15.0 L=5.0 nf=1	119.64	184.30 (54.0%)	119.38 (0.2%)
W=15.0 L=5.0 nf=2	210.58	260.49 (23.7%)	210.54 (0.0%)
W=15.0 L=5.0 nf=4	392.45	412.87 (5.2%)	392.86 (0.1%)
W=16.0 L=2.0 nf=1	72.03	94.35 (31.0%)	71.91 (0.2%)
W=16.0 L=2.0 nf=2	113.69	175.62 (54.5%)	113.96 (0.2%)
W=16.0 L=2.0 nf=4	197.00	338.15 (71.7%)	198.06 (0.5%)
W=20.0 L=5.0 nf=1	154.44	209.70 (35.8%)	154.10 (0.2%)
W=20.0 L=5.0 nf=2	271.83	311.28 (14.5%)	271.77 (0.0%)
W=20.0 L=5.0 nf=4	506.60	514.45 (1.5%)	507.13 (0.1%)
W=40.0 L=5.0 nf=1	293.64	311.28 (6.0%)	292.99 (0.2%)
W=40.0 L=5.0 nf=2	516.83	514.45 (0.5%)	516.73 (0.0%)
W=40.0 L=5.0 nf=4	963.20	920.78 (4.4%)	964.20 (0.1%)



Device: pfet_01v8_lvt

Parameters: ah=1.0000, bh=2.1900, aw=0.9971, bw=0.3176, cw=1.6414

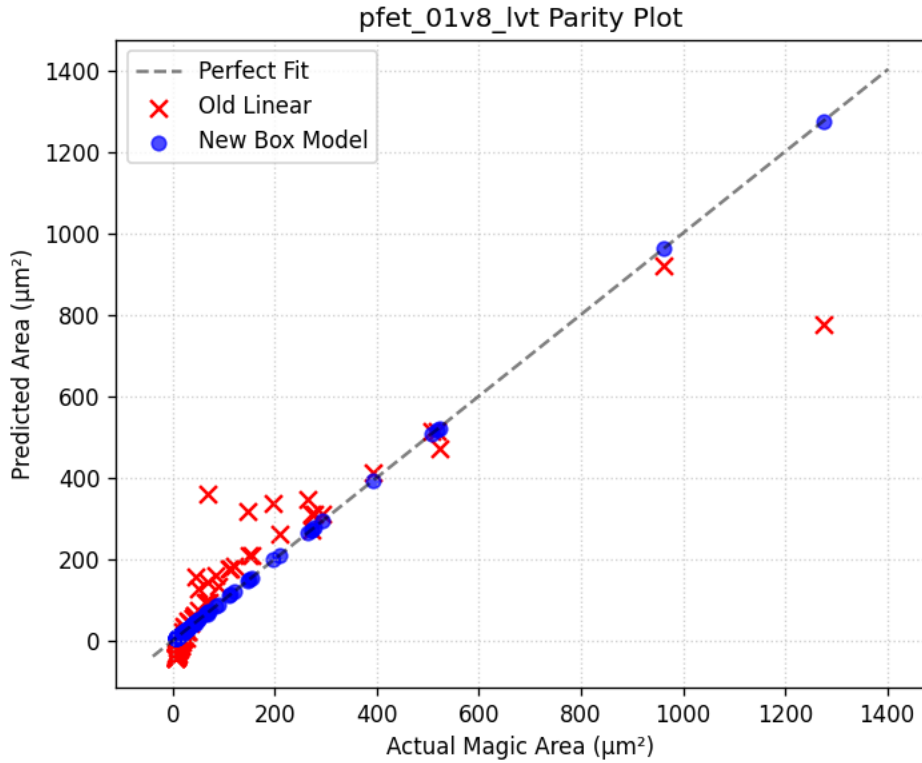
Height Model: $h_{um} = 1.0000 \cdot W + 2.1900$

Width Model: $w_{um} = 0.9971 \cdot (L \cdot nf) + 0.3176 \cdot nf + 1.6414$

Dimensions	Actual (μm ²)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	5.51	-43.39 (887.8%)	5.50 (0.1%)
W=0.42 L=0.15 nf=2	6.86	-41.25 (701.0%)	6.72 (2.1%)

W=0.42 L=0.15 nf=4	9.37	-36.99 (494.7%)	9.16 (2.2%)
W=0.42 L=0.5 nf=1	6.42	-32.30 (603.1%)	6.41 (0.1%)
W=0.42 L=1.0 nf=1	7.73	-16.46 (313.1%)	7.72 (0.1%)
W=0.45 L=0.15 nf=1	5.57	-43.23 (876.1%)	5.57 (0.1%)
W=0.45 L=0.15 nf=2	6.94	-40.95 (689.8%)	6.80 (2.1%)
W=0.45 L=0.15 nf=4	9.48	-36.38 (483.8%)	9.27 (2.2%)
W=0.6 L=0.15 nf=1	5.89	-42.47 (821.5%)	5.88 (0.1%)
W=0.6 L=0.15 nf=2	7.34	-39.43 (637.3%)	7.19 (2.1%)
W=0.6 L=0.15 nf=4	10.02	-33.33 (432.8%)	9.79 (2.2%)
W=1.0 L=0.15 nf=1	6.73	-40.44 (700.8%)	6.73 (0.1%)
W=1.0 L=0.15 nf=2	8.39	-35.36 (521.5%)	8.22 (2.1%)
W=1.0 L=0.5 nf=1	7.85	-29.35 (474.1%)	7.84 (0.1%)
W=1.0 L=0.5 nf=2	10.37	-24.27 (334.1%)	10.44 (0.7%)
W=1.0 L=0.5 nf=4	15.41	-14.12 (191.6%)	15.65 (1.6%)
W=1.0 L=2.0 nf=2	19.94	23.24 (16.6%)	19.99 (0.2%)
W=1.2 L=0.15 nf=1	7.15	-39.43 (651.2%)	7.15 (0.1%)
W=1.2 L=0.15 nf=2	8.92	-33.33 (473.8%)	8.73 (2.1%)
W=1.2 L=0.15 nf=4	12.17	-21.14 (273.7%)	11.90 (2.2%)
W=1.5 L=0.5 nf=1	9.08	-26.81 (395.4%)	9.07 (0.1%)
W=1.5 L=0.5 nf=2	11.99	-19.20 (260.1%)	12.08 (0.7%)
W=1.5 L=0.5 nf=4	17.82	-3.96 (122.2%)	18.10 (1.6%)
W=2.0 L=0.15 nf=4	15.04	-4.89 (132.5%)	14.71 (2.2%)
W=2.0 L=0.35 nf=2	12.36	-18.87 (252.6%)	12.46 (0.8%)
W=2.0 L=0.5 nf=1	10.31	-24.27 (335.5%)	10.30 (0.1%)
W=2.0 L=0.5 nf=2	13.62	-14.12 (203.7%)	13.72 (0.7%)
W=2.0 L=0.5 nf=4	20.24	6.20 (69.4%)	20.56 (1.6%)
W=2.0 L=1.0 nf=1	12.40	-8.44 (168.0%)	12.39 (0.1%)
W=2.0 L=1.0 nf=2	17.81	1.72 (90.3%)	17.89 (0.5%)
W=2.0 L=1.0 nf=4	28.62	22.04 (23.0%)	28.91 (1.0%)
W=2.0 L=5.0 nf=2	51.33	128.43 (150.2%)	51.32 (0.0%)
W=3.0 L=1.0 nf=1	15.36	-3.36 (121.8%)	15.34 (0.1%)
W=3.0 L=1.0 nf=2	22.06	11.88 (46.1%)	22.17 (0.5%)
W=3.0 L=1.0 nf=4	35.45	42.36 (19.5%)	35.81 (1.0%)
W=4.0 L=0.15 nf=4	22.22	35.75 (60.9%)	21.73 (2.2%)
W=4.0 L=0.5 nf=1	15.23	-14.12 (192.7%)	15.21 (0.1%)
W=4.0 L=0.5 nf=2	20.12	6.20 (69.2%)	20.26 (0.7%)
W=4.0 L=0.5 nf=4	29.90	46.83 (56.6%)	30.37 (1.6%)
W=4.0 L=1.0 nf=1	18.32	1.72 (90.6%)	18.30 (0.1%)
W=4.0 L=1.0 nf=2	26.31	22.04 (16.2%)	26.44 (0.5%)
W=4.0 L=1.0 nf=4	42.28	62.67 (48.2%)	42.71 (1.0%)
W=4.0 L=2.0 nf=1	24.51	33.40 (36.3%)	24.47 (0.2%)
W=4.0 L=2.0 nf=2	38.69	53.72 (38.8%)	38.78 (0.2%)
W=4.0 L=2.0 nf=4	67.04	94.35 (40.7%)	67.40 (0.5%)
W=4.0 L=5.0 nf=8	272.30	270.65 (0.6%)	272.77 (0.2%)
W=4.0 L=10.0 nf=4	265.12	347.77 (31.2%)	264.91 (0.1%)
W=6.0 L=2.0 nf=1	32.43	43.56 (34.3%)	32.38 (0.2%)
W=6.0 L=2.0 nf=2	51.19	74.03 (44.6%)	51.31 (0.2%)
W=6.0 L=2.0 nf=4	88.70	134.98 (52.2%)	89.18 (0.5%)

W=8.0 L=1.0 nf=1	30.16	22.04 (26.9%)	30.12 (0.1%)
W=8.0 L=1.0 nf=2	43.31	62.67 (44.7%)	43.52 (0.5%)
W=8.0 L=1.0 nf=4	69.60	143.94 (106.8%)	70.31 (1.0%)
W=8.0 L=2.0 nf=1	40.35	53.72 (33.1%)	40.28 (0.2%)
W=8.0 L=2.0 nf=2	63.69	94.35 (48.1%)	63.84 (0.2%)
W=8.0 L=2.0 nf=4	110.36	175.62 (59.1%)	110.95 (0.5%)
W=10.0 L=0.15 nf=1	25.72	5.27 (79.5%)	25.70 (0.1%)
W=10.0 L=0.15 nf=4	43.76	157.64 (260.2%)	42.79 (2.2%)
W=10.0 L=0.15 nf=8	67.17	360.81 (437.2%)	65.57 (2.4%)
W=10.0 L=5.0 nf=1	84.84	158.91 (87.3%)	84.65 (0.2%)
W=10.0 L=5.0 nf=2	149.33	209.70 (40.4%)	149.30 (0.0%)
W=10.0 L=5.0 nf=4	278.30	311.28 (11.9%)	278.59 (0.1%)
W=10.0 L=10.0 nf=1	145.79	317.30 (117.6%)	145.43 (0.3%)
W=10.0 L=10.0 nf=4	522.10	469.67 (10.0%)	521.68 (0.1%)
W=10.0 L=10.0 nf=10	1274.71	774.42 (39.2%)	1274.19 (0.0%)
W=15.0 L=5.0 nf=1	119.64	184.30 (54.0%)	119.38 (0.2%)
W=15.0 L=5.0 nf=2	210.58	260.49 (23.7%)	210.54 (0.0%)
W=15.0 L=5.0 nf=4	392.45	412.87 (5.2%)	392.86 (0.1%)
W=16.0 L=2.0 nf=1	72.03	94.35 (31.0%)	71.91 (0.2%)
W=16.0 L=2.0 nf=2	113.69	175.62 (54.5%)	113.96 (0.2%)
W=16.0 L=2.0 nf=4	197.00	338.15 (71.7%)	198.06 (0.5%)
W=20.0 L=5.0 nf=1	154.44	209.70 (35.8%)	154.10 (0.2%)
W=20.0 L=5.0 nf=2	271.83	311.28 (14.5%)	271.77 (0.0%)
W=20.0 L=5.0 nf=4	506.60	514.45 (1.5%)	507.13 (0.1%)
W=40.0 L=5.0 nf=1	293.64	311.28 (6.0%)	292.99 (0.2%)
W=40.0 L=5.0 nf=2	516.83	514.45 (0.5%)	516.73 (0.0%)
W=40.0 L=5.0 nf=4	963.20	920.78 (4.4%)	964.20 (0.1%)



Device: nfet_g5v0d10v5

Parameters: ah=1.0000, bh=2.5800, aw=1.0000, bw=0.2900, cw=2.0100

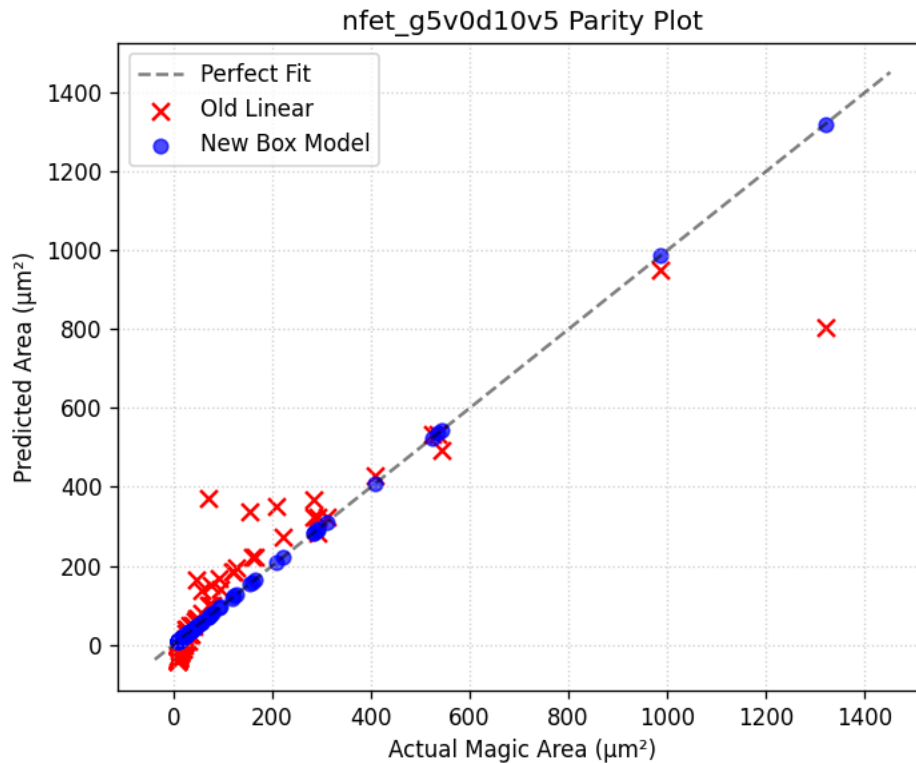
Height Model: $h_{um} = 1.0000 \cdot W + 2.5800$

Width Model: $w_{um} = 1.0000 \cdot (L \cdot nf) + 0.2900 \cdot nf + 2.0100$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	7.35	-42.04 (672.0%)	7.35 (0.0%)
W=0.42 L=0.15 nf=2	8.67	-39.86 (559.8%)	8.67 (0.0%)
W=0.42 L=0.15 nf=4	11.31	-35.50 (413.8%)	11.31 (0.0%)
W=0.42 L=0.5 nf=1	8.40	-30.39 (461.8%)	8.40 (0.0%)
W=0.42 L=1.0 nf=1	9.90	-13.75 (238.9%)	9.90 (0.0%)
W=0.45 L=0.15 nf=1	7.42	-41.89 (664.3%)	7.42 (0.0%)
W=0.45 L=0.15 nf=2	8.76	-39.55 (551.6%)	8.76 (0.0%)
W=0.45 L=0.15 nf=4	11.42	-34.87 (405.3%)	11.42 (0.0%)
W=0.6 L=0.15 nf=1	7.79	-41.11 (627.6%)	7.79 (0.0%)
W=0.6 L=0.15 nf=2	9.19	-37.99 (513.4%)	9.19 (0.0%)
W=0.6 L=0.15 nf=4	11.99	-31.76 (364.9%)	11.99 (0.0%)
W=1.0 L=0.15 nf=1	8.77	-39.03 (545.0%)	8.77 (0.0%)
W=1.0 L=0.15 nf=2	10.35	-33.83 (427.0%)	10.35 (0.0%)
W=1.0 L=0.5 nf=1	10.02	-27.38 (373.1%)	10.02 (0.0%)
W=1.0 L=0.5 nf=2	12.85	-22.18 (272.6%)	12.85 (0.0%)
W=1.0 L=0.5 nf=4	18.51	-11.79 (163.7%)	18.51 (0.0%)

W=1.0 L=2.0 nf=2	23.59	27.74 (17.6%)	23.59 (0.0%)
W=1.2 L=0.15 nf=1	9.26	-37.99 (510.2%)	9.26 (0.0%)
W=1.2 L=0.15 nf=2	10.92	-31.76 (390.7%)	10.92 (0.0%)
W=1.2 L=0.15 nf=4	14.25	-19.28 (235.3%)	14.25 (0.0%)
W=1.5 L=0.5 nf=1	11.42	-24.78 (316.9%)	11.42 (0.0%)
W=1.5 L=0.5 nf=2	14.65	-16.99 (216.0%)	14.65 (0.0%)
W=1.5 L=0.5 nf=4	21.09	-1.40 (106.6%)	21.09 (0.0%)
W=2.0 L=0.15 nf=4	17.27	-2.66 (115.4%)	17.27 (0.0%)
W=2.0 L=0.35 nf=2	15.07	-16.78 (211.4%)	15.07 (0.0%)
W=2.0 L=0.5 nf=1	12.82	-22.18 (273.0%)	12.82 (0.0%)
W=2.0 L=0.5 nf=2	16.44	-11.79 (171.7%)	16.44 (0.0%)
W=2.0 L=0.5 nf=4	23.68	8.99 (62.0%)	23.68 (0.0%)
W=2.0 L=1.0 nf=1	15.11	-5.54 (136.7%)	15.11 (0.0%)
W=2.0 L=1.0 nf=2	21.02	4.85 (76.9%)	21.02 (0.0%)
W=2.0 L=1.0 nf=4	32.84	25.64 (21.9%)	32.84 (0.0%)
W=2.0 L=5.0 nf=2	57.66	137.99 (139.3%)	57.66 (0.0%)
W=3.0 L=1.0 nf=1	18.41	-0.34 (101.9%)	18.41 (0.0%)
W=3.0 L=1.0 nf=2	25.61	15.24 (40.5%)	25.61 (0.0%)
W=3.0 L=1.0 nf=4	40.01	46.42 (16.0%)	40.01 (0.0%)
W=4.0 L=0.15 nf=4	24.81	38.91 (56.9%)	24.81 (0.0%)
W=4.0 L=0.5 nf=1	18.42	-11.79 (164.0%)	18.42 (0.0%)
W=4.0 L=0.5 nf=2	23.62	8.99 (61.9%)	23.62 (0.0%)
W=4.0 L=0.5 nf=4	34.02	50.56 (48.6%)	34.02 (0.0%)
W=4.0 L=1.0 nf=1	21.71	4.85 (77.7%)	21.71 (0.0%)
W=4.0 L=1.0 nf=2	30.20	25.64 (15.1%)	30.20 (0.0%)
W=4.0 L=1.0 nf=4	47.18	67.20 (42.4%)	47.18 (0.0%)
W=4.0 L=2.0 nf=1	28.29	38.14 (34.8%)	28.29 (0.0%)
W=4.0 L=2.0 nf=2	43.36	58.92 (35.9%)	43.36 (0.0%)
W=4.0 L=2.0 nf=4	73.50	100.49 (36.7%)	73.50 (0.0%)
W=4.0 L=5.0 nf=8	291.69	283.48 (2.8%)	291.69 (0.0%)
W=4.0 L=10.0 nf=4	284.06	366.77 (29.1%)	284.06 (0.0%)
W=6.0 L=2.0 nf=1	36.89	48.53 (31.5%)	36.89 (0.0%)
W=6.0 L=2.0 nf=2	56.54	79.71 (41.0%)	56.54 (0.0%)
W=6.0 L=2.0 nf=4	95.84	142.06 (48.2%)	95.84 (0.0%)
W=8.0 L=1.0 nf=1	34.91	25.64 (26.6%)	34.91 (0.0%)
W=8.0 L=1.0 nf=2	48.56	67.20 (38.4%)	48.56 (0.0%)
W=8.0 L=1.0 nf=4	75.86	150.34 (98.2%)	75.86 (0.0%)
W=8.0 L=2.0 nf=1	45.49	58.92 (29.5%)	45.49 (0.0%)
W=8.0 L=2.0 nf=2	69.72	100.49 (44.1%)	69.72 (0.0%)
W=8.0 L=2.0 nf=4	118.18	183.63 (55.4%)	118.18 (0.0%)
W=10.0 L=0.15 nf=1	30.82	7.74 (74.9%)	30.82 (0.0%)
W=10.0 L=0.15 nf=4	47.43	163.62 (245.0%)	47.43 (0.0%)
W=10.0 L=0.15 nf=8	69.57	371.46 (434.0%)	69.57 (0.0%)
W=10.0 L=5.0 nf=1	91.83	169.17 (84.2%)	91.83 (0.0%)
W=10.0 L=5.0 nf=2	158.38	221.13 (39.6%)	158.38 (0.0%)
W=10.0 L=5.0 nf=4	291.48	325.05 (11.5%)	291.48 (0.0%)
W=10.0 L=10.0 nf=1	154.73	335.60 (116.9%)	154.73 (0.0%)
W=10.0 L=10.0 nf=4	543.08	491.48 (9.5%)	543.08 (0.0%)

W=10.0 L=10.0 nf=10	1319.77	803.25 (39.1%)	1319.77 (0.0%)
W=15.0 L=5.0 nf=1	128.33	195.15 (52.1%)	128.33 (0.0%)
W=15.0 L=5.0 nf=2	221.33	273.09 (23.4%)	221.33 (0.0%)
W=15.0 L=5.0 nf=4	407.33	428.97 (5.3%)	407.33 (0.0%)
W=16.0 L=2.0 nf=1	79.89	100.49 (25.8%)	79.89 (0.0%)
W=16.0 L=2.0 nf=2	122.44	183.63 (50.0%)	122.44 (0.0%)
W=16.0 L=2.0 nf=4	207.54	349.90 (68.6%)	207.54 (0.0%)
W=20.0 L=5.0 nf=1	164.83	221.13 (34.2%)	164.83 (0.0%)
W=20.0 L=5.0 nf=2	284.28	325.05 (14.3%)	284.28 (0.0%)
W=20.0 L=5.0 nf=4	523.18	532.90 (1.9%)	523.18 (0.0%)
W=40.0 L=5.0 nf=1	310.83	325.05 (4.6%)	310.83 (0.0%)
W=40.0 L=5.0 nf=2	536.08	532.90 (0.6%)	536.08 (0.0%)
W=40.0 L=5.0 nf=4	986.58	948.59 (3.9%)	986.58 (0.0%)



Device: pfet_g5v0d10v5

Parameters: ah=1.0000, bh=2.9700, aw=1.0000, bw=0.2900, cw=2.2900

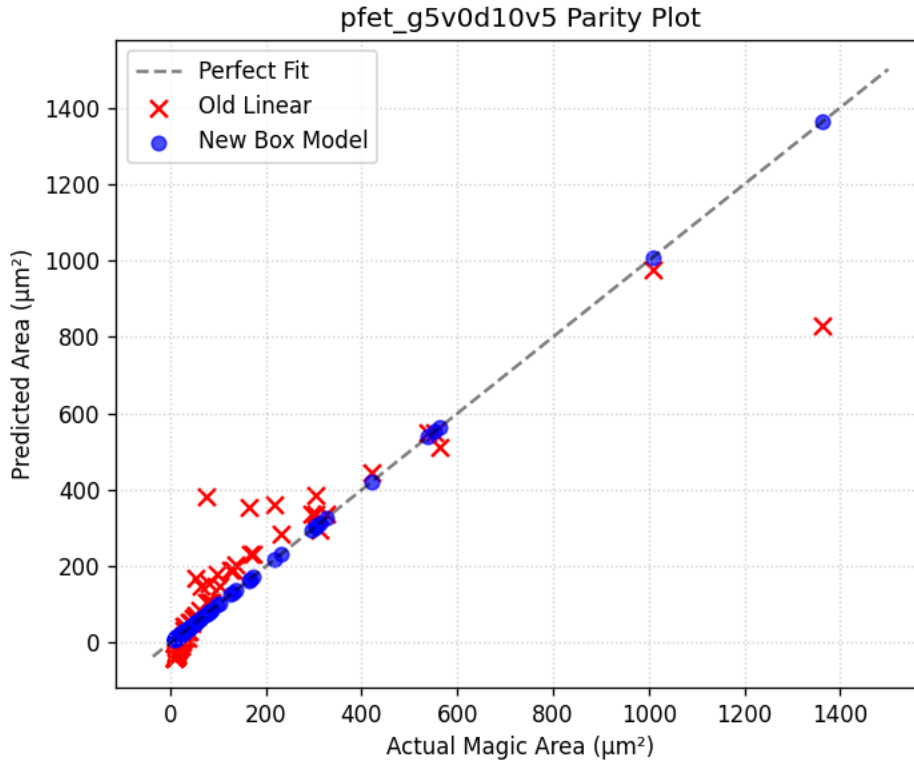
Height Model: $h_{\text{um}} = 1.0000 \cdot W + 2.9700$

Width Model: $w_{\text{um}} = 1.0000 \cdot (L \cdot \text{nf}) + 0.2900 \cdot \text{nf} + 2.2900$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	9.25	-40.57 (538.4%)	9.25 (0.0%)
W=0.42 L=0.15 nf=2	10.75	-38.34 (456.8%)	10.75 (0.0%)

W=0.42 L=0.15 nf=4	13.73	-33.88 (346.8%)	13.73 (0.0%)
W=0.42 L=0.5 nf=1	10.44	-28.40 (372.0%)	10.44 (0.0%)
W=0.42 L=1.0 nf=1	12.14	-11.02 (190.8%)	12.14 (0.0%)
W=0.45 L=0.15 nf=1	9.34	-40.41 (532.9%)	9.34 (0.0%)
W=0.45 L=0.15 nf=2	10.84	-38.03 (450.7%)	10.84 (0.0%)
W=0.45 L=0.15 nf=4	13.85	-33.25 (340.0%)	13.85 (0.0%)
W=0.6 L=0.15 nf=1	9.75	-39.62 (506.5%)	9.75 (0.0%)
W=0.6 L=0.15 nf=2	11.32	-36.43 (421.9%)	11.32 (0.0%)
W=0.6 L=0.15 nf=4	14.46	-30.06 (307.9%)	14.46 (0.0%)
W=1.0 L=0.15 nf=1	10.84	-37.49 (445.9%)	10.84 (0.0%)
W=1.0 L=0.15 nf=2	12.58	-32.19 (355.8%)	12.58 (0.0%)
W=1.0 L=0.5 nf=1	12.23	-25.32 (307.1%)	12.23 (0.0%)
W=1.0 L=0.5 nf=2	15.36	-20.02 (230.3%)	15.36 (0.0%)
W=1.0 L=0.5 nf=4	21.64	-9.40 (143.4%)	21.64 (0.0%)
W=1.0 L=2.0 nf=2	27.27	32.15 (17.9%)	27.27 (0.0%)
W=1.2 L=0.15 nf=1	11.38	-36.43 (420.0%)	11.38 (0.0%)
W=1.2 L=0.15 nf=2	13.22	-30.06 (327.4%)	13.22 (0.0%)
W=1.2 L=0.15 nf=4	16.89	-17.32 (202.6%)	16.89 (0.0%)
W=1.5 L=0.5 nf=1	13.77	-22.67 (264.7%)	13.77 (0.0%)
W=1.5 L=0.5 nf=2	17.30	-14.71 (185.0%)	17.30 (0.0%)
W=1.5 L=0.5 nf=4	24.36	1.22 (95.0%)	24.36 (0.0%)
W=2.0 L=0.15 nf=4	20.13	-0.34 (101.7%)	20.13 (0.0%)
W=2.0 L=0.35 nf=2	17.74	-14.61 (182.4%)	17.74 (0.0%)
W=2.0 L=0.5 nf=1	15.31	-20.02 (230.8%)	15.31 (0.0%)
W=2.0 L=0.5 nf=2	19.23	-9.40 (148.9%)	19.23 (0.0%)
W=2.0 L=0.5 nf=4	27.09	11.84 (56.3%)	27.09 (0.0%)
W=2.0 L=1.0 nf=1	17.79	-2.63 (114.8%)	17.79 (0.0%)
W=2.0 L=1.0 nf=2	24.20	7.99 (67.0%)	24.20 (0.0%)
W=2.0 L=1.0 nf=4	37.03	29.22 (21.1%)	37.03 (0.0%)
W=2.0 L=5.0 nf=2	63.96	147.08 (129.9%)	63.96 (0.0%)
W=3.0 L=1.0 nf=1	21.37	2.68 (87.5%)	21.37 (0.0%)
W=3.0 L=1.0 nf=2	29.07	18.61 (36.0%)	29.07 (0.0%)
W=3.0 L=1.0 nf=4	44.48	50.46 (13.4%)	44.48 (0.0%)
W=4.0 L=0.15 nf=4	28.23	42.13 (49.3%)	28.23 (0.0%)
W=4.0 L=0.5 nf=1	21.47	-9.40 (143.8%)	21.47 (0.0%)
W=4.0 L=0.5 nf=2	26.97	11.84 (56.1%)	26.97 (0.0%)
W=4.0 L=0.5 nf=4	37.99	54.30 (43.0%)	37.99 (0.0%)
W=4.0 L=1.0 nf=1	24.95	7.99 (68.0%)	24.95 (0.0%)
W=4.0 L=1.0 nf=2	33.94	29.22 (13.9%)	33.94 (0.0%)
W=4.0 L=1.0 nf=4	51.93	71.69 (38.1%)	51.93 (0.0%)
W=4.0 L=2.0 nf=1	31.92	42.76 (34.0%)	31.92 (0.0%)
W=4.0 L=2.0 nf=2	47.88	64.00 (33.6%)	47.88 (0.0%)
W=4.0 L=2.0 nf=4	79.81	106.46 (33.4%)	79.81 (0.0%)
W=4.0 L=5.0 nf=8	310.93	295.72 (4.9%)	310.93 (0.0%)
W=4.0 L=10.0 nf=4	302.85	384.65 (27.0%)	302.85 (0.0%)
W=6.0 L=2.0 nf=1	41.08	53.38 (29.9%)	41.08 (0.0%)
W=6.0 L=2.0 nf=2	61.62	85.23 (38.3%)	61.62 (0.0%)
W=6.0 L=2.0 nf=4	102.71	148.93 (45.0%)	102.71 (0.0%)

W=8.0 L=1.0 nf=1	39.27	29.22 (25.6%)	39.27 (0.0%)
W=8.0 L=1.0 nf=2	53.42	71.69 (34.2%)	53.42 (0.0%)
W=8.0 L=1.0 nf=4	81.73	156.62 (91.6%)	81.73 (0.0%)
W=8.0 L=2.0 nf=1	50.24	64.00 (27.4%)	50.24 (0.0%)
W=8.0 L=2.0 nf=2	75.36	106.46 (41.3%)	75.36 (0.0%)
W=8.0 L=2.0 nf=4	125.61	191.40 (52.4%)	125.61 (0.0%)
W=10.0 L=0.15 nf=1	35.41	10.28 (71.0%)	35.41 (0.0%)
W=10.0 L=0.15 nf=4	52.53	169.53 (222.7%)	52.53 (0.0%)
W=10.0 L=0.15 nf=8	75.36	381.87 (406.8%)	75.36 (0.0%)
W=10.0 L=5.0 nf=1	98.31	178.93 (82.0%)	98.31 (0.0%)
W=10.0 L=5.0 nf=2	166.92	232.02 (39.0%)	166.92 (0.0%)
W=10.0 L=5.0 nf=4	304.15	338.18 (11.2%)	304.15 (0.0%)
W=10.0 L=10.0 nf=1	163.16	352.80 (116.2%)	163.16 (0.0%)
W=10.0 L=10.0 nf=4	563.55	512.05 (9.1%)	563.55 (0.0%)
W=10.0 L=10.0 nf=10	1364.31	830.56 (39.1%)	1364.31 (0.0%)
W=15.0 L=5.0 nf=1	136.21	205.48 (50.8%)	136.21 (0.0%)
W=15.0 L=5.0 nf=2	231.27	285.10 (23.3%)	231.27 (0.0%)
W=15.0 L=5.0 nf=4	421.40	444.35 (5.4%)	421.40 (0.0%)
W=16.0 L=2.0 nf=1	86.88	106.46 (22.5%)	86.88 (0.0%)
W=16.0 L=2.0 nf=2	130.32	191.40 (46.9%)	130.32 (0.0%)
W=16.0 L=2.0 nf=4	217.21	361.26 (66.3%)	217.21 (0.0%)
W=20.0 L=5.0 nf=1	174.11	232.02 (33.3%)	174.11 (0.0%)
W=20.0 L=5.0 nf=2	295.62	338.18 (14.4%)	295.62 (0.0%)
W=20.0 L=5.0 nf=4	538.65	550.52 (2.2%)	538.65 (0.0%)
W=40.0 L=5.0 nf=1	325.71	338.18 (3.8%)	325.71 (0.0%)
W=40.0 L=5.0 nf=2	553.02	550.52 (0.5%)	553.02 (0.0%)
W=40.0 L=5.0 nf=4	1007.65	975.19 (3.2%)	1007.65 (0.0%)



Device: nfet_g5v0d16v0

Parameters: ah=1.0000, bh=3.9800, aw=1.0000, bw=2.3139, cw=3.6741

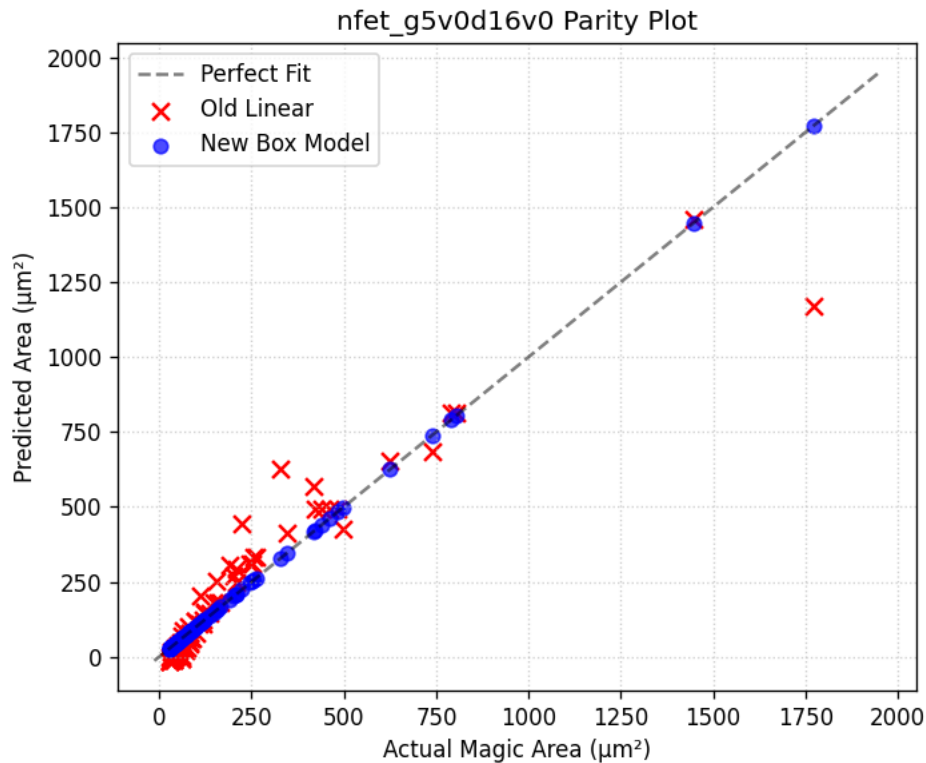
Height Model: $h_{um} = 1.0000 \cdot W + 3.9800$

Width Model: $w_{um} = 1.0000 \cdot (L \cdot nf) + 2.3139 \cdot nf + 3.6741$

Dimensions	Actual (μm^2)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	27.02	-14.95 (155.3%)	27.01 (0.0%)
W=0.42 L=0.15 nf=2	37.84	-11.57 (130.6%)	37.85 (0.0%)
W=0.42 L=0.15 nf=4	59.53	-4.81 (108.1%)	59.53 (0.0%)
W=0.42 L=0.5 nf=1	28.56	-1.35 (104.7%)	28.55 (0.0%)
W=0.42 L=1.0 nf=1	30.76	18.07 (41.2%)	30.75 (0.0%)
W=0.45 L=0.15 nf=1	27.20	-14.71 (154.1%)	27.19 (0.0%)
W=0.45 L=0.15 nf=2	38.10	-11.09 (129.1%)	38.11 (0.0%)
W=0.45 L=0.15 nf=4	59.94	-3.85 (106.4%)	59.94 (0.0%)
W=0.6 L=0.15 nf=1	28.12	-13.50 (148.0%)	28.11 (0.0%)
W=0.6 L=0.15 nf=2	39.39	-8.67 (122.0%)	39.40 (0.0%)
W=0.6 L=0.15 nf=4	61.97	0.98 (98.4%)	61.97 (0.0%)
W=1.0 L=0.15 nf=1	30.58	-10.28 (133.6%)	30.57 (0.0%)
W=1.0 L=0.15 nf=2	42.83	-2.24 (105.2%)	42.84 (0.0%)
W=1.0 L=0.5 nf=1	32.32	3.32 (89.7%)	32.31 (0.0%)
W=1.0 L=0.5 nf=2	46.31	11.36 (75.5%)	46.32 (0.0%)
W=1.0 L=0.5 nf=4	74.35	27.46 (63.1%)	74.35 (0.0%)

W=1.0 L=2.0 nf=2	61.25	69.65 (13.7%)	61.26 (0.0%)
W=1.2 L=0.15 nf=1	31.81	-8.67 (127.3%)	31.79 (0.0%)
W=1.2 L=0.15 nf=2	44.55	0.98 (97.8%)	44.56 (0.0%)
W=1.2 L=0.15 nf=4	70.09	20.30 (71.0%)	70.08 (0.0%)
W=1.5 L=0.5 nf=1	35.57	7.34 (79.4%)	35.55 (0.0%)
W=1.5 L=0.5 nf=2	50.96	19.41 (61.9%)	50.97 (0.0%)
W=1.5 L=0.5 nf=4	81.82	43.56 (46.8%)	81.81 (0.0%)
W=2.0 L=0.15 nf=4	80.91	46.06 (43.1%)	80.91 (0.0%)
W=2.0 L=0.35 nf=2	53.82	21.63 (59.8%)	53.83 (0.0%)
W=2.0 L=0.5 nf=1	38.81	11.36 (70.7%)	38.80 (0.0%)
W=2.0 L=0.5 nf=2	55.61	27.46 (50.6%)	55.63 (0.0%)
W=2.0 L=0.5 nf=4	89.28	59.66 (33.2%)	89.28 (0.0%)
W=2.0 L=1.0 nf=1	41.80	30.79 (26.3%)	41.79 (0.0%)
W=2.0 L=1.0 nf=2	61.59	46.89 (23.9%)	61.61 (0.0%)
W=2.0 L=1.0 nf=4	101.24	79.08 (21.9%)	101.24 (0.0%)
W=2.0 L=5.0 nf=2	109.43	202.31 (84.9%)	109.45 (0.0%)
W=3.0 L=1.0 nf=1	48.79	38.84 (20.4%)	48.78 (0.0%)
W=3.0 L=1.0 nf=2	71.89	62.99 (12.4%)	71.91 (0.0%)
W=3.0 L=1.0 nf=4	118.17	111.28 (5.8%)	118.17 (0.0%)
W=4.0 L=0.15 nf=4	107.97	110.45 (2.3%)	107.97 (0.0%)
W=4.0 L=0.5 nf=1	51.79	27.46 (47.0%)	51.77 (0.0%)
W=4.0 L=0.5 nf=2	74.21	59.66 (19.6%)	74.23 (0.0%)
W=4.0 L=0.5 nf=4	119.14	124.05 (4.1%)	119.14 (0.0%)
W=4.0 L=1.0 nf=1	55.78	46.89 (15.9%)	55.76 (0.0%)
W=4.0 L=1.0 nf=2	82.19	79.08 (3.8%)	82.21 (0.0%)
W=4.0 L=1.0 nf=4	135.10	143.48 (6.2%)	135.10 (0.0%)
W=4.0 L=2.0 nf=1	63.76	85.75 (34.5%)	63.74 (0.0%)
W=4.0 L=2.0 nf=2	98.15	117.94 (20.2%)	98.17 (0.0%)
W=4.0 L=2.0 nf=4	167.02	182.33 (9.2%)	167.02 (0.0%)
W=4.0 L=5.0 nf=8	496.28	427.68 (13.8%)	496.24 (0.0%)
W=4.0 L=10.0 nf=4	422.38	493.18 (16.8%)	422.38 (0.0%)
W=6.0 L=2.0 nf=1	79.74	101.84 (27.7%)	79.72 (0.0%)
W=6.0 L=2.0 nf=2	122.75	150.14 (22.3%)	122.77 (0.0%)
W=6.0 L=2.0 nf=4	208.88	246.72 (18.1%)	208.88 (0.0%)
W=8.0 L=1.0 nf=1	83.74	79.08 (5.6%)	83.72 (0.0%)
W=8.0 L=1.0 nf=2	123.39	143.48 (16.3%)	123.42 (0.0%)
W=8.0 L=1.0 nf=4	202.82	272.26 (34.2%)	202.82 (0.0%)
W=8.0 L=2.0 nf=1	95.72	117.94 (23.2%)	95.70 (0.0%)
W=8.0 L=2.0 nf=2	147.35	182.33 (23.7%)	147.38 (0.0%)
W=8.0 L=2.0 nf=4	250.74	311.11 (24.1%)	250.74 (0.0%)
W=10.0 L=0.15 nf=1	85.84	62.16 (27.6%)	85.81 (0.0%)
W=10.0 L=0.15 nf=4	189.15	303.62 (60.5%)	189.15 (0.0%)
W=10.0 L=0.15 nf=8	326.99	625.57 (91.3%)	326.93 (0.0%)
W=10.0 L=5.0 nf=1	153.64	250.61 (63.1%)	153.61 (0.0%)
W=10.0 L=5.0 nf=2	255.83	331.09 (29.4%)	255.86 (0.0%)
W=10.0 L=5.0 nf=4	460.36	492.07 (6.9%)	460.36 (0.0%)
W=10.0 L=10.0 nf=1	223.54	444.88 (99.0%)	223.51 (0.0%)
W=10.0 L=10.0 nf=4	739.96	686.35 (7.2%)	739.96 (0.0%)

W=10.0 L=10.0 nf=10	1772.94	1169.28 (34.0%)	1772.85 (0.0%)
W=15.0 L=5.0 nf=1	208.59	290.85 (39.4%)	208.55 (0.0%)
W=15.0 L=5.0 nf=2	347.33	411.58 (18.5%)	347.37 (0.0%)
W=15.0 L=5.0 nf=4	625.01	653.05 (4.5%)	625.01 (0.0%)
W=16.0 L=2.0 nf=1	159.64	182.33 (14.2%)	159.60 (0.0%)
W=16.0 L=2.0 nf=2	245.75	311.11 (26.6%)	245.79 (0.0%)
W=16.0 L=2.0 nf=4	418.18	568.68 (36.0%)	418.18 (0.0%)
W=20.0 L=5.0 nf=1	263.54	331.09 (25.6%)	263.49 (0.0%)
W=20.0 L=5.0 nf=2	438.83	492.07 (12.1%)	438.88 (0.0%)
W=20.0 L=5.0 nf=4	789.66	814.02 (3.1%)	789.65 (0.0%)
W=40.0 L=5.0 nf=1	483.34	492.07 (1.8%)	483.25 (0.0%)
W=40.0 L=5.0 nf=2	804.83	814.02 (1.1%)	804.92 (0.0%)
W=40.0 L=5.0 nf=4	1448.26	1457.93 (0.7%)	1448.25 (0.0%)



Device: pfet_g5v0d16v0

Parameters: ah=1.0000, bh=4.2800, aw=1.0000, bw=1.9200, cw=3.9700

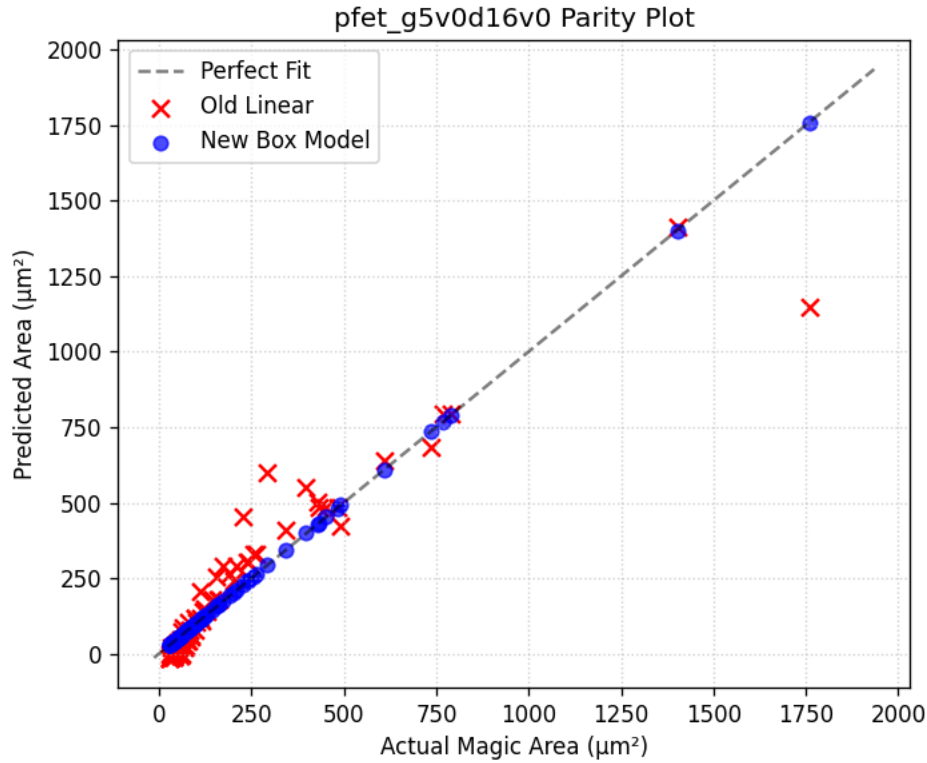
Height Model: $h_{um} = 1.0000 \cdot W + 4.2800$

Width Model: $w_{um} = 1.0000 \cdot (L \cdot nf) + 1.9200 \cdot nf + 3.9700$

Dimensions	Actual (μm ²)	Old Model Err	New Model Err
W=0.42 L=0.15 nf=1	28.39	-14.86 (152.4%)	28.39 (0.0%)
W=0.42 L=0.15 nf=2	38.12	-11.62 (130.5%)	38.12 (0.0%)

W=0.42 L=0.15 nf=4	57.58	-5.13 (108.9%)	57.58 (0.0%)
W=0.42 L=0.5 nf=1	30.03	-0.84 (102.8%)	30.03 (0.0%)
W=0.42 L=1.0 nf=1	32.38	19.20 (40.7%)	32.38 (0.0%)
W=0.45 L=0.15 nf=1	28.57	-14.63 (151.2%)	28.57 (0.0%)
W=0.45 L=0.15 nf=2	38.36	-11.16 (129.1%)	38.36 (0.0%)
W=0.45 L=0.15 nf=4	57.94	-4.21 (107.3%)	57.94 (0.0%)
W=0.6 L=0.15 nf=1	29.48	-13.47 (145.7%)	29.48 (0.0%)
W=0.6 L=0.15 nf=2	39.58	-8.84 (122.3%)	39.58 (0.0%)
W=0.6 L=0.15 nf=4	59.78	0.42 (99.3%)	59.78 (0.0%)
W=1.0 L=0.15 nf=1	31.89	-10.38 (132.6%)	31.89 (0.0%)
W=1.0 L=0.15 nf=2	42.82	-2.66 (106.2%)	42.82 (0.0%)
W=1.0 L=0.5 nf=1	33.74	3.64 (89.2%)	33.74 (0.0%)
W=1.0 L=0.5 nf=2	46.52	11.36 (75.6%)	46.52 (0.0%)
W=1.0 L=0.5 nf=4	72.07	26.80 (62.8%)	72.07 (0.0%)
W=1.0 L=2.0 nf=2	62.36	71.46 (14.6%)	62.36 (0.0%)
W=1.2 L=0.15 nf=1	33.10	-8.84 (126.7%)	33.10 (0.0%)
W=1.2 L=0.15 nf=2	44.44	0.42 (99.0%)	44.44 (0.0%)
W=1.2 L=0.15 nf=4	67.13	18.95 (71.8%)	67.13 (0.0%)
W=1.5 L=0.5 nf=1	36.93	7.50 (79.7%)	36.93 (0.0%)
W=1.5 L=0.5 nf=2	50.92	19.08 (62.5%)	50.92 (0.0%)
W=1.5 L=0.5 nf=4	78.90	42.24 (46.5%)	78.90 (0.0%)
W=2.0 L=0.15 nf=4	76.93	43.66 (43.3%)	76.93 (0.0%)
W=2.0 L=0.35 nf=2	53.44	20.79 (61.1%)	53.44 (0.0%)
W=2.0 L=0.5 nf=1	40.13	11.36 (71.7%)	40.13 (0.0%)
W=2.0 L=0.5 nf=2	55.33	26.80 (51.6%)	55.33 (0.0%)
W=2.0 L=0.5 nf=4	85.72	57.68 (32.7%)	85.72 (0.0%)
W=2.0 L=1.0 nf=1	43.27	31.39 (27.4%)	43.27 (0.0%)
W=2.0 L=1.0 nf=2	61.61	46.83 (24.0%)	61.61 (0.0%)
W=2.0 L=1.0 nf=4	98.28	77.72 (20.9%)	98.28 (0.0%)
W=2.0 L=5.0 nf=2	111.85	207.11 (85.2%)	111.85 (0.0%)
W=3.0 L=1.0 nf=1	50.16	39.11 (22.0%)	50.16 (0.0%)
W=3.0 L=1.0 nf=2	71.42	62.27 (12.8%)	71.42 (0.0%)
W=3.0 L=1.0 nf=4	113.93	108.60 (4.7%)	113.93 (0.0%)
W=4.0 L=0.15 nf=4	101.43	105.42 (3.9%)	101.43 (0.0%)
W=4.0 L=0.5 nf=1	52.91	26.80 (49.3%)	52.91 (0.0%)
W=4.0 L=0.5 nf=2	72.95	57.68 (20.9%)	72.95 (0.0%)
W=4.0 L=0.5 nf=4	113.02	119.44 (5.7%)	113.02 (0.0%)
W=4.0 L=1.0 nf=1	57.05	46.83 (17.9%)	57.05 (0.0%)
W=4.0 L=1.0 nf=2	81.23	77.72 (4.3%)	81.23 (0.0%)
W=4.0 L=1.0 nf=4	129.58	139.48 (7.6%)	129.58 (0.0%)
W=4.0 L=2.0 nf=1	65.33	86.90 (33.0%)	65.33 (0.0%)
W=4.0 L=2.0 nf=2	97.79	117.78 (20.4%)	97.79 (0.0%)
W=4.0 L=2.0 nf=4	162.70	179.55 (10.4%)	162.70 (0.0%)
W=4.0 L=5.0 nf=8	491.25	423.27 (13.8%)	491.25 (0.0%)
W=4.0 L=10.0 nf=4	427.66	500.10 (16.9%)	427.66 (0.0%)
W=6.0 L=2.0 nf=1	81.11	102.34 (26.2%)	81.11 (0.0%)
W=6.0 L=2.0 nf=2	121.41	148.66 (22.5%)	121.41 (0.0%)
W=6.0 L=2.0 nf=4	202.00	241.31 (19.5%)	202.00 (0.0%)

W=8.0 L=1.0 nf=1	84.61	77.72 (8.1%)	84.61 (0.0%)
W=8.0 L=1.0 nf=2	120.47	139.48 (15.8%)	120.47 (0.0%)
W=8.0 L=1.0 nf=4	192.18	263.00 (36.8%)	192.18 (0.0%)
W=8.0 L=2.0 nf=1	96.89	117.78 (21.6%)	96.89 (0.0%)
W=8.0 L=2.0 nf=2	145.03	179.55 (23.8%)	145.03 (0.0%)
W=8.0 L=2.0 nf=4	241.30	303.07 (25.6%)	241.30 (0.0%)
W=10.0 L=0.15 nf=1	86.25	59.10 (31.5%)	86.25 (0.0%)
W=10.0 L=0.15 nf=4	174.93	290.70 (66.2%)	174.93 (0.0%)
W=10.0 L=0.15 nf=8	293.17	599.51 (104.5%)	293.17 (0.0%)
W=10.0 L=5.0 nf=1	155.51	253.43 (63.0%)	155.51 (0.0%)
W=10.0 L=5.0 nf=2	254.33	330.63 (30.0%)	254.33 (0.0%)
W=10.0 L=5.0 nf=4	451.96	485.04 (7.3%)	451.96 (0.0%)
W=10.0 L=10.0 nf=1	226.91	453.78 (100.0%)	226.91 (0.0%)
W=10.0 L=10.0 nf=4	737.56	685.38 (7.1%)	737.56 (0.0%)
W=10.0 L=10.0 nf=10	1758.87	1148.59 (34.7%)	1758.87 (0.0%)
W=15.0 L=5.0 nf=1	209.96	292.03 (39.1%)	209.96 (0.0%)
W=15.0 L=5.0 nf=2	343.38	407.83 (18.8%)	343.38 (0.0%)
W=15.0 L=5.0 nf=4	610.21	639.44 (4.8%)	610.21 (0.0%)
W=16.0 L=2.0 nf=1	160.01	179.55 (12.2%)	160.01 (0.0%)
W=16.0 L=2.0 nf=2	239.51	303.07 (26.5%)	239.51 (0.0%)
W=16.0 L=2.0 nf=4	398.50	550.11 (38.0%)	398.50 (0.0%)
W=20.0 L=5.0 nf=1	264.41	330.63 (25.0%)	264.41 (0.0%)
W=20.0 L=5.0 nf=2	432.43	485.04 (12.2%)	432.43 (0.0%)
W=20.0 L=5.0 nf=4	768.46	793.84 (3.3%)	768.46 (0.0%)
W=40.0 L=5.0 nf=1	482.21	485.04 (0.6%)	482.21 (0.0%)
W=40.0 L=5.0 nf=2	788.63	793.84 (0.7%)	788.63 (0.0%)
W=40.0 L=5.0 nf=4	1401.46	1411.45 (0.7%)	1401.46 (0.0%)



2. Poly Resistors

Poly resistor areas are modeled using a linear relationship with length (L) for a given fixed width (W). Each specific width has its own slope and intercept:

$$\text{Area} = \text{slope} * L + \text{intercept}$$

Device: res_high_po_0p35

Fixed Width (W)	Slope	Intercept
0.35	2.0100	11.6982

Device: res_high_po_0p69

Fixed Width (W)	Slope	Intercept
0.69	2.3500	13.6770

Device: res_high_po_1p41

Fixed Width (W)	Slope	Intercept
1.41	3.0700	17.8674

Device: res_high_po_2p85

Fixed Width (W)	Slope	Intercept
2.85	4.5100	26.2482

Device: res_high_po_5p73

Fixed Width (W)	Slope	Intercept
5.73	7.3900	43.0098

Device: res_xhigh_po_0p35

Fixed Width (W)	Slope	Intercept
0.35	2.0100	11.6982

Device: res_xhigh_po_0p69

Fixed Width (W)	Slope	Intercept
0.69	2.3500	13.6770

Device: res_xhigh_po_1p41

Fixed Width (W)	Slope	Intercept
1.41	3.0700	17.8674

Device: res_xhigh_po_2p85

Fixed Width (W)	Slope	Intercept
2.85	4.5100	26.2482

Device: res_xhigh_po_5p73

Fixed Width (W)	Slope	Intercept
5.73	7.3900	43.0098

Device: res_generic_po

Fixed Width (W)	Slope	Intercept
0.5	2.1600	12.8736
1.0	2.6600	15.8536
2.0	3.6600	21.8136

Device: res_generic_nd

Fixed Width (W)	Slope	Intercept
0.5	2.0800	5.4288

1.0	2.5800	6.7338
2.0	3.5800	9.3438

Device: res_generic_pd

Fixed Width (W)	Slope	Intercept
0.5	2.0800	5.4288
1.0	2.5800	6.7338
2.0	3.5800	9.3438

3. Capacitors

MIM Capacitors

MIM capacitors are modeled as a rectangle with a uniform border overhead (b) around the drawn dimensions:

$$\text{Area} = (W + 2b) * (L + 2b)$$

- cap_mim_m3_1: border overhead (b) = 0.6209 μm

- cap_mim_m3_2: border overhead (b) = 1.0484 μm

Varactor Capacitors

Varactors fall back to nearest-neighbor lookup because their scaling is highly non-linear depending on exact array layout.

- cap_var_lvt: 3 points in database for nearest-neighbor.

- cap_var_hvt: 3 points in database for nearest-neighbor.

4. Bipolar Junction Transistors (BJTs)

BJTs are fixed-size layout macros. Their areas are static.

Device	Width (μm)	Height (μm)	Area (μm^2)
npn_05v5_W1p00L1p00	8.6200	8.6200	74.3044
npn_05v5_W1p00L2p00	8.6200	9.6200	82.9244
pnnp_05v5_W0p68L0p68	3.9800	3.9800	15.8404
pnnp_05v5_W3p40L3p40	6.7000	6.7000	44.8900